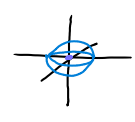
Ej 1:

$|E|$ y $|H|$, 10 Km → Campo lejano ($\beta r >> 1$)

$P_{in} = 15KW$, 70% eficiencia , dipolo de Hertz → 

→ $P_{rad} = 0,7 \cdot P_{in} = 10,5 \, KW$ (70 W)

Pot. prom. → $\langle P \rangle = \frac{P_{rad}}{4\pi r^2} = \frac{10,5KW}{4\pi \cdot (10 \, 10^3)^2} = 8,35 \cdot 10^6 \frac{W}{m} = 8,35 \frac{\mu W}{m}$

→ $|\langle P \rangle| = \frac{|E|^2}{2|Z_0|}$ → $Z_0 = Z_{00} = 120\pi$, asumo vacío, sin pérdidas.

$8,35 \cdot 10^{-6} = \frac{E_0^2}{2 \cdot 377}$ → $E_0 = \sqrt{8,35 \, 10^{-6} \cdot 2 \cdot 377} = 0,079 \frac{V}{m} = 7,9 \frac{mV}{m}$

$H_0 = \frac{|E_0|}{|Z_{00}|} = \frac{0,079}{377} = 2,1 \cdot 10^{-4} = 0,21 \frac{mA}{m}$

Ej 2:

$G_{max} = 20$ $W = P_{Rad} = 100 W$

1) Pot. irrad. equiv. (PIRE)

$G = \eta \cdot D$, $\eta = \frac{P_{Rad}}{P_{in}}$, $D = \frac{P_{rad}}{P_0}$

Necesito P_0 para igualar antena (PIRE)

→ $P_{IRE} = 20 \cdot 100W = 2000 W$

2) $\langle P \rangle$ a 15Km de la antena?

→ Irradia $P_{Rad} = 2000W$ = Antena a analizar

$\frac{2000}{4\pi \cdot 15 \cdot 10^3} = 7,07 \cdot 10^{-7} \frac{W}{m^2} = \langle P \rangle_{15Km}$

3) 2000 W

Ej 3:

Dipolo → $I_0 = 20A$, $\Delta l = \frac{\lambda}{50}$

P_{Max} a 1Km ? R_{Rad} ?

Para Dipolo Corto:

Pot. Radiada

$P_{Rad} = I_0^2 \, 40 \pi^2 \left(\frac{\Delta l}{\lambda}\right)$ → $20^2 \cdot 40 \cdot \pi^2 \cdot \left(\frac{\lambda/50}{\lambda}\right) = 63 W$

Resist. de Rad.

$R_{rad} = 80\pi^2 \left(\frac{\Delta l}{\lambda}\right)^2 = 80 \, \pi^2 \cdot \left(\frac{1}{50}\right)^2 = 0,316 \, \Omega = 316 \, m\Omega$

Ej 4:

Dipolo , $\Delta l = 1m$, $I_0 = 12A$, $f = 1MHz$

$\langle P \rangle$ en 5Km a 45° ?

$\langle P \rangle = \frac{I_0^2 \, \Delta l^2 \, sen^2\theta \, \beta^2 \, Z_{00}}{32 r^2 \pi^2}$ ✓

$\beta = \omega \sqrt{\mu\varepsilon} = 2\pi \cdot 1 \cdot 10^6 \sqrt{\mu\varepsilon}$

$\beta = 0,021 \frac{rad}{m}$

$= \frac{12^2 \cdot 1 \cdot sen^2(45°) \cdot (0,021)^2 \cdot 377}{32 \cdot (5 \cdot 10^3)^2 \cdot \pi^2}$

$\langle P \rangle_{@45° \atop 5Km} = 1,516 \cdot 10^{-9} \simeq 1,5 \frac{nW}{m^2}$

Ej 5:

Dipolo , $\Delta l = 2m$, $f = 1MHz$, cobre y radio = 1mm

1) $\sigma_{cobre} = 5,8 \cdot 10^7 \frac{S}{m}$ → $\rho = 1,72 \cdot 10^{-8} \frac{\Omega}{m}$

Opcion 1 (R como siempre):

$R_{loss} = \rho \cdot \frac{\Delta l}{A}$, $A = \pi \cdot r^2 = \pi \cdot (1 \cdot 10^{-3})^2 = \pi \cdot 10^{-6}$

$R_{loss} = 1,72 \cdot 10^{-8} \cdot 2 \cdot \frac{1}{\pi} \cdot 10^6 = 0,0109 \, \Omega$

$R_{rad} = 80\pi^2 \left(\frac{\Delta l}{\lambda}\right)^2$, $\Delta l = 2$ y $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{\mu\varepsilon}} = \frac{1}{f\sqrt{\mu\varepsilon}} = 300 \, m$

→ $R_{Rad} = 80 \, \pi^2 \left(\frac{2}{300}\right)^2 = 0,0350 \, \Omega$

→ $\eta = \frac{R_{Rad}}{R_{Rad} + R_{loss}} = \frac{0,035}{0,0459} \simeq 0,765 \rightsquigarrow 76,5\%$

Opcion 2 (R con ef. skin):

$R_{loss} = \frac{l}{2\pi r \delta \sigma}$, $\delta = \sqrt{\frac{1}{\pi f \mu \sigma}}$ (Efecto skin)

$\delta = 6,6 \cdot 10^{-5}$ → $R_{loss} = 0,083 \, \Omega$

$R_{Rad} = 0,035$

→ $\eta = \frac{0,035}{0,035 + 0,083} = 0,296 \rightsquigarrow 29,6\%$

2) $D_{dipolo} = 1,5$, $\eta = 29,6\%$

$G = 0,30 \cdot 1,5 = 0,441 = -3,56 \, dBi$

3) $P_{Rad} = I_0^2 \cdot 40\pi^2 \left(\frac{\Delta l}{\lambda}\right)$

- $P_{Rad} = 80W$ si se refiere solo a la antena

4) - $\frac{80}{\eta} = 270W$ si se refiere a P_{in}

→ 270 : $I_0 = \sqrt{\frac{270}{40\pi^2 \cdot \left(\frac{1}{150}\right)^2}} = 124 \, A$

Ej 6:

$\eta = 90\%$, $D = 7 dBi = 5$

$G = \eta \cdot D = 0,9 \cdot 5 = 4,51 = 6,54 \, dBi$

Ej 7:

A) Dipolo media onda → $l = \frac{\lambda}{2}$, $P_{rad} = 1KW$ $f = 50MHz$

B) $G_R = 3dBi$ a $x = 30 \, Km$
$= 2$

Friis → $P_{Rec_B} = P_{Rad_A} \cdot G_A \cdot G_B \cdot \left(\frac{\lambda}{4\pi x}\right)^2$

G_A → Dipolo de media onda → 2,15 dBi

$G_A = 1,64$

$\lambda = \frac{2\pi}{\beta} = \frac{1}{f\sqrt{\mu\varepsilon}} = 6 \, m$

→ $P_{Rec_B} = 1KW \cdot 2 \cdot 1,64 \cdot \left(\frac{6}{4\pi \, 30 \cdot 10^3}\right)^2$

$P_{Rec_B} = 8,3 \cdot 10^{-7} W$

Ej 8: → Duplicado de Ej 7.

Ej 9:

2 Dipolos $\frac{\lambda}{2}$, $x = 150m$, $P_{Rad_A} = 2KW$, $f = 300MHz$

$G = 1,64$

Friis → $P_{Rec_B} = P_{Rad_A} \cdot G_A \cdot G_B \cdot \left(\frac{\lambda}{4\pi \cdot x}\right)^2$, $\lambda = \frac{3 \, 10^8}{300 \, 10^6} = 1m$

$P_{Rec_B} = 2 \cdot 10^3 \cdot (1,64)^2 \cdot \left(\frac{1}{4\pi \cdot 150}\right)^2$

$P_{Rec_B} = 1,51 \, mW$

Ej 10:

1) A) $G_A = 55 dBi$ @ 14 GHz , $x = 36500 \, Km = 36,5 \cdot 10^6 m$ ($= 316,2 \cdot 10^3$)

B) $G_B = 35 dBi$ ($= 3,16 \cdot 10^3$) , $P_{Rec_{Bmin}} = 8 \, pW = 8 \cdot 10^{-12} W$

→ $P_{Rec_B} = P_{Rad_A} \cdot G_A \cdot G_B \cdot \left(\frac{\lambda}{4\pi x}\right)^2$ ($= 2,1 \cdot 10^{-21}$) , $\lambda = \frac{3 \cdot 10^8}{14 \cdot 10^9} m = 0,021 \, m = 21 \, \mu m$

→ $\frac{8 \cdot 10^{-12}}{3,16 \cdot 316 \cdot 10^3 \cdot 10^3 \cdot 2,1 \cdot 10^{-21}} = P_{Rad_A}$

$P_{Rad_{Amin}} = 3,81 \, W$

2) ROE = 1,1

$1,1 = \frac{1 + |\Gamma|}{1 - |\Gamma|}$ → $1,1 - 1,1|\Gamma| = 1 + |\Gamma|$

→ $1,1 - 1 = |\Gamma|(1 + 1,1)$

$\frac{0,1}{2,1} = |\Gamma| = 0,048$

→ $P_{Rec_B} = P_{Rad_A} \cdot (1 - |\Gamma|^2)^2 \cdot G_A \cdot G_B \cdot \left(\frac{\lambda}{4\pi x}\right)^2$ //

3) $-20 dB = 0,01$ → $P_{Rad_A} \cdot \frac{1}{0,01} = P_{Rad_A}$ Necesario para contrarrestar la atenuacion.